[This question paper contains 8 printed pages.]

Your Roll No.....

Sr. No. of Question Paper : 3377 **H**

Unique Paper Code : 42347610

Name of the Paper : Computer Networks (DSE)

Name of the Course : **B.Sc. (Programme)**

Semester : VI

Duration : 3 Hours Maximum Marks : 75

Instructions for Candidates

1. Write your Roll No. on the top immediately on receipt of this question paper.

2. The paper has **two** sections.

3. **All** questions in **'Section A'** are compulsory.

4. Attempt any **five** questions from **'Section B'**.

5. Parts of a question must be answered together.

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P.T.O.

Section A

(Compulsory)

1. (a) Consider the following data fragment in the middle of a data stream in a frame : (2)

A B ESC C ESC FLAG FLAG D

Give the output when byte-stuffing technique is used for framing.

(b) Name the layers on which the following networking devices operate : (2)

(i) Bridge

(ii) Router

(iii) Gateway

(iv) Hub

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(c) Indicate whether the following employ connection-less or connection-oriented mechanism : (2)

(i) WhatsApp call

(ii) Facebook chat

(iii) Postal System

(iv) WhatsApp Message

(d) Describe the function of Address Resolution Protocol (ARP). In which layer does it operate? (2)

(e) List at least two differences between the internet and the extranet. (2)

2. (a) Differentiate between the star and bus topologies. List one benefit and one drawback for each. (3)

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(b) What is the difference between a single-bit error, a two-bit error, and a burst error? (3)

(c) Differentiate between circuit switching and packet switching. (3)

(d) What is purpose of cladding in an optical fiber cable? Explain with the help of a diagram. (3)

(e) What do understand by simplex, half duplex and full duplex mode of data communication? Give one example of each. (3)

Section B

(*Attempt any* **Five**)

3. (a) We want to transmit the bit-string "10110011" using Cyclic Redundancy Check (CRC). The generator bit-string is "1001". Obtain the transmitted bit-string. (4)

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(b) What is the purpose of File Transfer Protocol (FTP)? Briefly explain the different transmission modes of FTP. (6)

4. (a) What do you understand by Domain Name System (DNS)? Which protocol does it use to resolve the name mapping? Discuss how domain name structure is organized. (6)

(b) What are the components of a URL? And what punctuation is used to separate the components? (4)

5. (a) List the differences between ISO-OSI and TCP/IP protocol stacks. (4)

(b) Find the class and subnet mask for each of the following IP addresses. (6)

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(i) 208.34.54.12

(ii) 238.34.2.1

(iii) 114.34.2.8

6. (a) List the protocols that are used by modern E-mail Application system. Briefly discuss the role played by these protocols in the E-mail system. (6)

(b) A system has an n-layer protocol hierarchy. Applications generate messages of length M bytes. At each of the layers, an h-byte header is added. What fraction of the network bandwidth is filled with headers? (4)

7. (a) Briefly discuss the functionality of each layer of the ISO-OSI model. (7)

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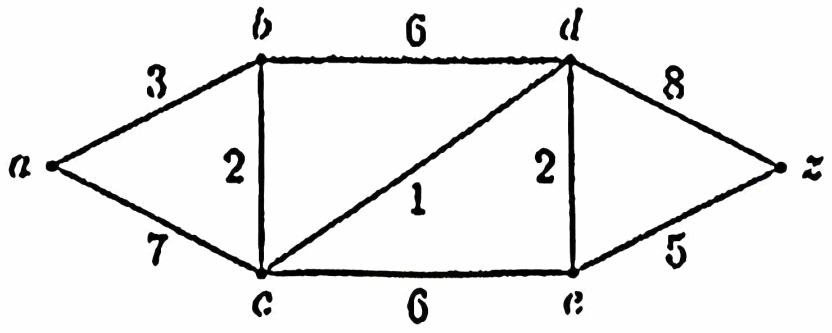
(b) Classify computer networks according to the coverage area. (3)

8. (a) What is the difference between multicast and broadcast in terms of network traffic? Explain with suitable examples. (4)

(b) Find the length and shortest path between "a" and "z" in the following weighted graph. (6)



9. Write short notes on the following **(any two)** :

(a) Stop-and-Wait Protocol

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(b) TELNET

(c) Radio Waves (10)

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